1 How the SMOTE Literature Uses Data

A reader who wants to compare two SMOTE variants, or to choose a benchmark for a new one, needs answers to questions that the papers rarely answer together. Can the numbers that two papers report on a dataset of the same name be compared? How broad is the evidence behind the benchmarks the field relies on? How difficult, beyond the imbalance ratio, are the datasets that can be checked? When a paper reports that its method beats SMOTE, on what kind of data does that happen, and can a reader check it? This section answers these questions from the 1,654 dataset entries and 835 comparisons with SMOTE (834 per dataset and one average over datasets) that we extracted from the 76 papers of our corpus. Three of them (ENN, AdaBoost and one review) contain no evaluation on data, so counts below use the 73 papers with an evaluation on data; Tomek links is kept because it illustrates its method on one synthetic two-class set.

The central observation is that the evidence for a new oversampler is only as interpretable as the description of its data, and in this literature that description is often incomplete: the same name can denote different tasks, modifications are frequently undeclared, and a sizeable share of papers report results that cannot be traced back to individual datasets.

1.1 Extraction Procedure

For every paper we read the full text and any supplementary file in our collection, and recorded each dataset as printed, including spelling, with its group in the paper, source repository, size, number of features and classes, minority and majority counts, and the imbalance ratio where the paper prints them, together with the table and page they come from. For each dataset we also recorded how it reached the experiment: *original* (used as distributed, with its native classes), *repository-defined* (a two-class split published by a repository such as KEEL or imbalanced-learn, used as is), *modified* (the authors chose the minority classes, merged or dropped classes, set a threshold, undersampled or injected noise), or *synthetic* (generated by the authors). Where a paper did not state its source, two fixed rules were applied and flagged: names that follow the KEEL file convention (for example `ecoli-0-1_vs_5`, `yeast4`) were labelled repository-defined, and common names in multiclass papers with no stated change were labelled original. Two further rules were fixed before counting: a set that is two-class in its source and is used without comment was labelled original, and a set from which the authors removed rows (for example, records with missing values) or attributes was labelled modified.

For the results we copied, per dataset, the printed score of the proposed method and of SMOTE from one results table selected by a fixed rule: G-mean, else AUC, else F-measure, else accuracy; the classifier highlighted in the abstract, else the first reported; the unperturbed condition. We recorded values rather than wins so that any definition of a win can be applied later. Plain SMOTE was distinguished from SMOTE embedded in a multiclass or ensemble wrapper (Static-SMOTE, OVO-SMOTE [FERNANDEZ2013ANALYSING], SMOTE-all, SMOTE+AdaBoost). Values printed in a paper are never overwritten; corrections against the source repository are stored beside them with their justification. Every extracted value was then checked against the PDF a second time: a value was accepted when the dataset name and its numbers appear on the same line of the cited page, and every table that is printed as an image or whose text layer is garbled was read from the rendered page instead. All 1,654 dataset entries and 835 comparisons passed this check.

Evidence: ai-draft/full/SCHEMA.md; build/full_extraction.py; KEEL and multiclass rules approved by Toon 18 Oct, rules A and B 18 Oct; 454 extraction notes in out/full_uncertain.csv, each resolved in out/uncertain_resolution.csv; second check: build/verify_pdf.py, out/visual_checks.json, out/verified_*.csv; values also checked by Toon against the PDFs (28 Oct 2026).

1.2 The Benchmark Landscape

The 73 papers with an evaluation on data list 1,654 dataset entries. One of them, the single toy example in Tomek links, is a description rather than a name and is set aside here, leaving 1,653 named entries. These reduce to 771 distinct names once names are lower-cased, text in parentheses is dropped, and underscores and spaces are treated alike. Treating spellings that differ only in punctuation as one dataset (`page-blocks`, `pageblocks` and the singular `page-block`) leaves 680 datasets, and grouping each numbered or split version with its main set (for example `yeast4` or `glass-0-1-6_vs_2` with yeast and glass), while keeping non-standard names such as `glass identification` apart, leaves 459 source families. Most datasets appear once: 463 of the 680 occur in a single paper. Counted as datasets, ecoli appears in 29 papers, yeast in 22 and glass in 19; counted with their splits, the ecoli family appears in 42 papers, the yeast family in 39 and the glass family in 35, about half of the 73. Half of all uses of a family fall on 37 of the 459 families. Names whose printed number of features contradicts their main set are kept in its family when they are the same source with a different composition (for example `abalone8`, `oil1`) and kept apart when they are a different data set that shares the prefix (for example `car1`, a microarray set with 9,182 features).

Splits cut from one source are not copies of each other. Among the verifiable two-class KEEL files (Section 1.5), the 14 yeast splits range in imbalance ratio from 2.5 to 41.4 and in noise index from 0.29 to 0.86, and the 19 ecoli and 13 glass splits from 0.12 to 0.69 and from 0.24 to 0.77. They do share the source's attributes (7 or 8 for yeast, 6 or 7 for ecoli, 9 for glass), its domain and often its examples: `yeast4` and `yeast-0-5-6-7-9_vs_4` use the same 51 minority examples and differ only in which classes form the majority.

Outcomes on splits of one source are also related. In the eleven papers that compare with plain SMOTE on several splits of the same source (282 per-dataset comparisons, 262 same-family pairs; families by the rule in the evidence note), whether the proposed method wins does not cluster by family (pairs agree on win, tie or loss in 0.60 of cases, as under random relabelling), but the size of its margin does: the within-paper ranks of the margin differ by 0.29 on average for two splits of one source against 0.34 when family labels are permuted within each paper ($p = 0.005$, 10,000 permutations; leaving out one paper at a time gives p between 0.002 and 0.048).

Splits cut from one source vary the task (its imbalance ratio and where the minority examples lie) but not the source (its domain, attributes and dimensionality), and within the papers that evaluate several splits of one source, the margin over SMOTE tends to be slightly more alike within a family than between families. Papers should therefore report the number of source families alongside the number of datasets, and should not treat each split as independent evidence.

What the reader should do. Report the source family of each dataset alongside the count, so that readers can see how many independent sources stand behind a claim, and keep splits of one source together when testing for significance.

Evidence: out/verified_datasets.csv; counts reproduced by build/reconcile_counts.py (out/reconcile_counts.json): rule R1 sets aside the Tomek links placeholder (row 1558, kept in the file), R2 normalises with build/analyse_dataset_usage.py norm(), R3 gives exact-name, dataset and family counts under rule T1 (build/family_rule.py, out/dataset_families.csv; the earlier prefix count gave 39/39/36 and is superseded); datasets and families per name: out/landscape_families.csv; concentration by dataset and family: out/benchmark_concentration.json. With the placeholder the counts are 772 and 547; removing all punctuation would leave 678 names. The 605 names of the 6 Oct list are superseded by the 771 of the full re-read. Spread within the yeast, ecoli and glass families: difficulty/family_spread.py → family_spread.json. Datasets and families everywhere in this subsection: rule T1 (Toon, 28 Oct 2026; names differing only by punctuation are one dataset, numbered or split versions belong to their main set, non-standard names are not merged),

Table 1: Data Sets Considered as Benchmarks for Cross-Family Comparison. F: used by ten or more papers; D: used by three to nine papers, every printed size at least 10,000 instances; V: needs verification. Papers count the dataset (spellings that differ only in punctuation are one dataset); family papers include the numbered and split versions of the dataset. Values are as printed and differ between papers. [a]Four printed versions, not pooled.

Dataset	Group	Papers	Family papers	Instances	Size printed	Classes	Modified
`ecoli`	F	29	42	301–336	25/31	2, 4, 5, 8	8/31
`page-blocks`	F	22	34	545–5,476	17/22	2, 4, 5	6/22
`yeast`	F	22	39	483–1,485	16/22	2, 6, 8, 9, 10	4/22
`pima`	F	21	21	590–769	19/23	2	2/23
`vehicle`	F	21	30	713–940	19/23	2, 3, 4	10/23
`haberman`	F	20	20	252–306	19/22	2	2/22
`glass`	F	19	35	167–214	18/21	2, 6	7/21
`balance`	F	16	16	625	14/16	3	3/16
`wine`	F	16	16	130–178	15/18	2, 3	6/18
`satimage`	F	15	16	6,430–6,435	11/15	2, 6, 7	8/15
`letter`	D	9	9	11,057–20,000	9/9	2, 26	6/9
`nursery`	D	7	7	12,690–12,960	7/7	2, 4, 5	3/7
`mammography`	D	5	5	11,183	4/5	2	0/5
`adult`	D	3	3	32,562–48,842	2/3	2	1/3
`avila`	D	3	4	10,430–20,850	3/3	2, 11	1/3
`default`	D	3	3	30,000	2/3	2	0/3
`shoppers`	D	3	3	12,330–12,331	3/3	2	0/3
`shuttle`	V	7	15	2,175–57,999[a]	6/7	2, 5	1/7

build/family_rule.py → out/dataset_families.csv; outcomes linked to records through out/outcome_dataset_links.csv; build/family_dependence.py → out/family_dependence.json (key t1).

Candidate benchmarks for a cross-family comparison. Table 1 lists 18 datasets considered as benchmarks for comparing method families: the 11 datasets we set out to examine and the datasets that meet one of two written rules. The first rule takes every dataset used by at least ten papers (33 datasets). The second takes datasets used by three to nine papers whose every printed size is at least 10,000 instances, a size that no entry of the first group reaches (7 datasets). The groups follow from these two rules on use and printed size only and say nothing about overlap, noise, sparsity, feature type or any other property. IR is not used, because its definition differs between papers. No dataset used by three or more papers is printed with 100 or more features every time it is reported; this is a property of the rule, not evidence that the corpus contains no high-dimensional data sets.

Family counts follow the same rule for every dataset. The dataset `shuttle` is printed with four different sizes (2,175, 14,500, 43,500 and 57,999 instances) and once without a size. Two pairs of papers print matching sizes and class counts; the other versions are single reports, and OVO-SMOTE prints five classes in one table and seven class counts in another. The versions are therefore kept apart and not pooled.

The frequently used benchmarks are small: among the entries that print a size, half have 800 or fewer instances and half have nine or fewer features. Names printed with 10,000 or more instances in every paper are used by three to nine papers each, so a comparison across method families that wants to cover data size has to look beyond the frequent set.

Table 2: How the Papers Choose and Obtain Their Data. Top: paper level, 73 papers with an evaluation on data. Bottom: dataset entries.

Reason given for the choice of datasets (papers)	
Own reason (diversity, difficulty, application, . . .)	48
Cites prior work	14
"Widely used" only	3
Not stated	8
Benchmark list adopted from earlier work, declared (papers)	20 (wholesale 11)
Binarised, minority rule not stated (papers)	8 of 37
How each dataset entry reached the experiment	
Original	534
Repository-defined (KEEL, imbalanced-learn)	470
Modified by the authors	331
Synthetic	120
Cannot be told from the paper	199

Limits of the evidence. Size, dimensionality and the number of classes are printed for 80.2%, 75.9% and 77.4% of the 1,653 named entries, and class proportions in some form for 56.8%. A measured class overlap exists for 36 entries in two papers, on two incompatible scales, and the curated evidence contains no statement about sparsity. What the papers print therefore cannot show which ranges of overlap, noise or sparsity the benchmarks cover; the candidates in Table 1 are chosen on use and size only. Section 1.5 measures these properties directly for the entries whose files can be identified.

Evidence: out/benchmark_candidates.csv and .json (build/landscape_candidates.py; rules and approval status in `criteria`); table rows written by build/landscape_manuscript.py; out/landscape_families.csv (446 groups, 82 flagged and left unmerged); out/shuttle_versions.csv (build/shuttle_versions.py; OVO-SMOTE Table 2 and Table 3, p.6, checked 19 Oct); out/characteristics_coverage.json; out/benchmark_concentration.json. All records are browsable in the Atlas (manuscript tab and Landscape tab).

1.3 How Datasets Are Chosen and Modified

Table 2 shows that modification by the authors is common: 331 of 1,654 dataset entries were changed by the authors themselves, and for 199 the paper does not let a reader tell how the data were obtained. Eleven papers give no reason for their choice of datasets or only call them widely used.

Modification is not a fault in itself: binarising a multiclass set or raising the imbalance ratio are legitimate ways to probe a method. What matters for the reader is whether the modification is declared where a reader looks for it. A KEEL file name such as `yeast-0-5-6-7-9_vs_4` declares its split; a paper that prints only "Yeast" after choosing its own minority classes does not.

[Toon] This is where your axis "declared vs undeclared" belongs. Evidence for where each paper declares its modification exists for the 7 papers re-read on 16 Oct and in how_obtained_detail of the full re-read.

Evidence: out/data_handling_summary.json (paper-level codes); out/full_summary.json how_obtained_all (entry level, with rules).

Table 3: Imbalance-Ratio Definitions Printed in the Papers

Definition	Formula	Used by (examples)
Largest over smallest class	$N_{\max}/N_{\min}$	MDO [ABDI2016MDO], CBH-SID [PALLI2022CBHSID], MKC-SMOTE [WANG2024MKC] (its IR)
Majority over minority (binary wording)	$N_{\mathrm{maj}}/N_{\min}$	OVO-SMOTE (defined for the binary case, applied to multiclass sets)
Percentage minority	$100\,N_{\min}/N$	RUSBoost [SEIFFERT2010RUSBOOST]
Per-class imbalance degree	$\sum_{q\neq i} N_q/((C-1)N_i)$	OREM-M [ZHU2023OREM] (overall IR only cited, not printed)
Summed pairwise multiclass IR	$\sum_{i<j}(N_i/N_j - 1)$	SMOM [ZHU2017SMOM], GDHS [YAN2025GDHS], MKC-SMOTE (IR_{overall})
Imbalance degree Id	distance from uniform [ORTIGOSAHERNANDEZ2017IMBALANCE]	SHSampler [ZHENG2022SHSAMPLER] (cited, no formula printed)
Minimum class prior	$N_{\min}/N$	DSRBF [FERNANDEZNAVARRO2011DSRBF]
Coefficient of variation	$\mathrm{sd}(N_i)/\mathrm{mean}(N_i)$	COG-OS [WU2007COGOS]
Proportion pair	e.g. 0.04:0.96	MWMOTE (wording: minority to majority)
Not stated (values equal $N_{\max}/N_{\min}$)	n/a	MC-CCR [KOZIARSKI2020MCCCR], MC-RBO [KRAWCZYK2020MCRBO]

1.4 The Same Name Is Not the Same Data

Of the 192 datasets that at least two papers print with a size (spellings that differ only in punctuation counted as one dataset), 90 appear with more than one size: thyroid with 7,200 instances in five papers and 720 in four, dermatology with 358 (the KEEL version) in eight and 366 (UCI) in two, ecoli with 336, 327, 310 and 301, and pima with 768, 769, 635 and 590. Some of these differences are declared subsamples; the name alone does not tell which. MWMOTE [BARUA2014MWMOTE] illustrates the problem inside a single paper: its Satimage (6,435 instances) and Statlandsat (4,435) are the full Statlog Landsat set and its training file, counted as two of its twenty datasets, and its common names are ambiguous in both directions (its Ecoli matches the KEEL file `ecoli1` exactly, while its Yeast matches no published split).

Imbalance ratios add a second layer. Table 3 lists the definitions found in the corpus; some are not convertible into one another, and some papers print IR without a formula.

Errors also travel with shared benchmark tables: MC-CCR and MC-RBO both print an IR of 1.17 for vehicle, a value no definition produces from its class counts.

What the reader should do. Compare results across papers only after mapping each name to its exact file and class split, and report the IR formula together with its values.

Evidence: out/verified_datasets.csv: names normalised, printed n_instances compared across papers (192 datasets, rule T1, with a printed size in two or more papers; 198 if spellings are kept apart; out/reconcile_counts.json size_variation); the earlier 20 of 75 from ai-draft/dataset_variants.json covered 8 papers only and is superseded; mwmote-table5-audit.md; extraction-notes.md; full re-read ir_definition per paper in out/full_summary.json.

Table 4: Difficulty Profile of the Verifiable Tasks. A task can carry several flags; entries count how often the corpus uses a task.

	Binary and binarised		One-vs-rest
	Tasks	Entries	Tasks
All	113	512	171
Small disjuncts (rare $\geq 20\%$)	11 (10%)	39 (8%)	7 (4%)
Outliers ($> 20\%$)	24 (21%)	120 (23%)	14 (8%)
Small sample ($N_{\min}/d < 10$)	79 (70%)	345 (67%)	70 (41%)
None of the three	34 (30%)	167 (33%)	97 (57%)
Imbalance ratio > 9	68	284	90
Noise index, median [P25, P75]	0.40 [0.25, 0.72]		0.16 [0.03, 0.47]

1.5 How Difficult Are the Datasets That Can Be Checked?

The papers rarely describe how hard their data are beyond the imbalance ratio. A numerical overlap measure is printed for 36 of the 1,653 named entries, all in two papers, and noise or borderline examples are described for 5; three of 72 papers report such a property of their original data (Section 1.2). We therefore measured difficulty ourselves on the entries whose data can be identified. An entry counts as verifiable when its name matches a public file and the printed number of instances (and of minority examples, when printed) equals that file's, or when the paper prints no size but the name matches a KEEL file and the authors did not modify the data. This holds for 643 entries from 61 papers, which use 131 files. Entries modified by their authors without printed characteristics (28) are left out, because the data actually used cannot be reconstructed.

Each two-class file, and each binarisation that a paper describes precisely enough to reproduce, defines one task: 113 tasks, used by 512 entries in 41 papers. One-vs-rest tasks that we built from multiclass files (171) are not tasks any paper evaluated and are reported apart. Every minority example is labelled safe, borderline, rare or outlier from the classes of its five nearest neighbours [napierala2016types], with distances measured by HVDM [wilson1997improved]. The noise index of a task is the share of minority examples that are borderline or outliers; we report it as a continuous value because we found no published threshold for it. Three flags use published thresholds: small disjuncts when at least 20% of minority examples are rare and outliers when more than 20% are outliers [napierala2016types], and small sample when there are fewer than ten minority examples per feature [jain2000statistical]. The imbalance ratio $N_{\text{maj}}/N_{\min}$, computed from the files, is grouped into the bands 1.5 to 3, 3 to 9 and above 9 [fernandez2008study] and is not used as a difficulty flag.

Table 4 shows that the most common property of the binary tasks is a small sample: 79 of 113 have fewer than ten minority examples per feature. The flags combine in only four profiles: small sample alone (55 tasks), small sample with outliers (13), all three flags (11) and none (34); every task with small disjuncts also carries the other two flags. The noise index separates tasks that the flags do not (Figure 1a): among the 34 tasks without a flag it ranges from 0 to 0.84, with ten tasks at 0.5 or above (haberman 0.84, bupa 0.79, vehicle1 0.74, yeast1 0.72 among them) and eight below 0.1. Its median is 0.37, 0.40, 0.31 and 0.50 in the imbalance-ratio bands up to 1.5 (4 tasks), 1.5 to 3 (22), 3 to 9 (19) and above 9 (68), and its rank correlation with the imbalance ratio is 0.15.

The measurable part of the corpus is also low-dimensional (Figure 1b). The printed number of features has a median of 9 and half of the entries lie between 4 and 10 features. The 59 entries with more than 100 features come from 54 names in 18 papers, no name is used by more than two papers, and none of them could be verified; the verified tasks have at most 85 features (median 9).

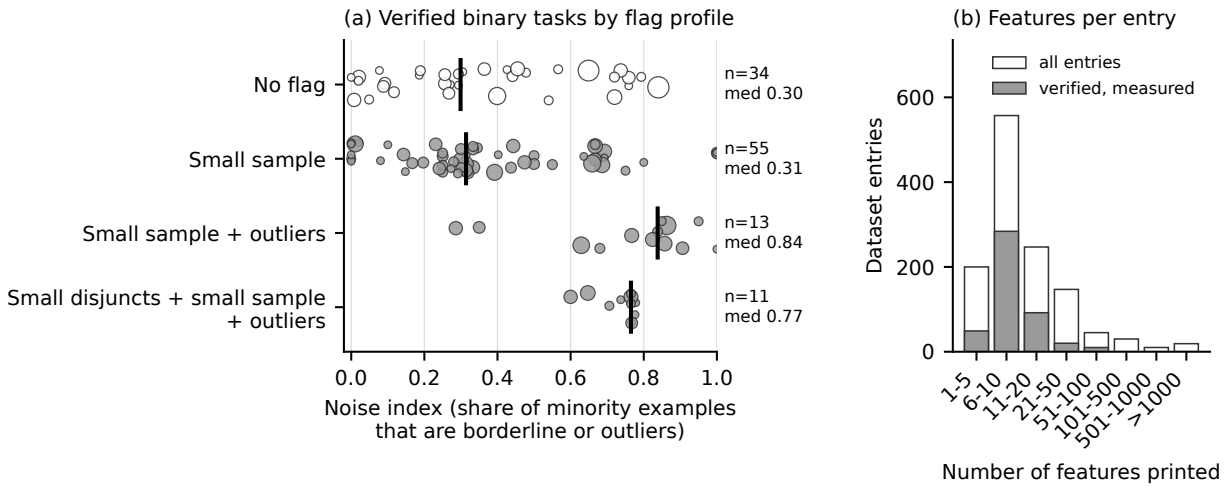


Figure 1: (a) Noise index of the 113 verifiable binary and binarised tasks, one row per flag profile; marker area grows with the number of entries that use the task, and the vertical bar marks the median. (b) Number of features printed for the 1,255 entries that report it, with the entries that could be verified and measured. The bins are for display only.

The imbalance ratio, the one difficulty-related number most papers print, says little about where the minority examples lie: tasks with a low ratio can be as mixed as tasks with a high one, and a task without any flag is not necessarily easy. Conclusions drawn from the common benchmarks speak mainly for small, low-dimensional tables with few minority examples per feature.

[**Toon**] Write the interpretation in your own words. Decide whether the difficulty profile should be reported per paper (which papers evaluate only on flag-free or only on small-sample tasks) once the win/loss analysis resumes.

Limits of the measurement. The files come from mirrors of the repositories (the keel-ds and imbalanced-databases packages and PMLB); for the 72 KEEL files in the official KEEL table, instance and feature counts match in every case, and the remaining files have not yet been compared with the repositories. PMLB stores nominal attributes as integer codes, so they enter HVDM as numeric attributes. The labels are computed in the full feature space, not in any two-dimensional projection.

Evidence: difficulty/verifiability_entries.csv (643 verifiable entries, 61 papers, 131 files); difficulty/measure.py (HVDM, $k = 5$, Napierala types incl. the rule for rare examples) → difficulty_measures.csv; group_cited.py → difficulty_groups_cited.csv; profile_final.py → profile_final_summary.json (Table 4); fig_noise_summary.json (rank correlation); dimension_profile.json; keel_official_check.csv; figure: difficulty/fig_manuscript.py; reported properties: out/characteristics_coverage.json (Analysis E).

1.6 Where Reported Wins over SMOTE Come From

At the paper level, 44 of the 73 papers use plain SMOTE as a baseline, 13 use SMOTE inside a wrapper, and 16 do not compare with SMOTE at all. Fewer papers give rows that can be compared: 30 papers print per-dataset scores for both the proposed method and plain SMOTE in the same table, giving 541 comparisons (Table 5). In these, the proposed method wins in 420 rows, ties in

Table 5: Per-Dataset Comparisons of the Proposed Method with Plain SMOTE, by How the Dataset Was Obtained. Wins, ties and losses compare printed values under interim rule A (one metric fixed per paper; equal at the printed precision is a tie); scores are not comparable across papers. KNNOR's single row, a mean over its 19 datasets, is not a per-dataset comparison and is reported in the text instead.

How obtained	Rows	Papers	Proposed wins	SMOTE wins	Ties
Original	166	26	132	29	5
Repository-defined	119	9	90	13	16
Modified	143	14	109	30	4
Synthetic	5	1	5	0	0
Cannot be told	105	13	83	9	13
Result row not matched to a dataset	3	2	1	1	1
All	541	30	420	82	39

39 and loses in 82. KNNOR [ISLAM2022KNNOR] prints only a mean over its 19 datasets (G-score .8363 for KNNOR and .8238 for SMOTE); we report it here as an aggregate result and do not count it as a per-dataset comparison.

Comparisons with SMOTE inside a wrapper are kept apart and are not pooled with Table 5: all 13 wrapper papers print per-dataset pairs, giving 293 rows. In one of them, RB-Boost [DIEZPASTOR2015RANDOMBALANCE], the extracted pairs do not meet the same-table rule. The scores of its RB-Boost come from the Boosting-family table of its supplement (Supplementary Table 11) and those of its SMOTE ensemble (ESM) from the data-processing-family table (Supplementary Table 3). Both tables report G-mean under one protocol (J48 with Laplace smoothing, ensembles of 100, stratified 5×2-fold cross-validation), but the paper compares methods only within each family and never compares RB-Boost with ESM directly. Its 86 rows, 63 of them on repository-defined data, are kept in our data for audit and excluded from every comparison set reported here; the other 12 wrapper papers contribute 207 rows.

Split by data type, 29 of the 30 papers include at least one standard dataset (original or repository-defined) in the comparison, and on these 285 standard rows SMOTE wins in 42 and ties in 21. One paper reports its per-dataset comparison with SMOTE only on modified, synthetic or unidentifiable data (INOS [CAO2013INOS]); KNNOR, whose only comparison is the aggregate above, is not counted here. Conversely, six papers compare only on standard data (GDO [XIE2022GDO], MC-MBRC [MA2024MCMBRC], MKC-SMOTE, OREM-M, SCUT [AGRAWAL2015SCUT], SPE [LIU2020SPE]).

[Toon] Your hypothesis H1 lives here. Look at the shares of "SMOTE higher" per row of Table 5 and at the per-paper split in the Atlas (fourth tab, wins over SMOTE by data type). What you conclude, and how strongly, is yours; the within-paper comparison (standard vs modified rows of the same paper) is the strongest test because it holds the paper constant.

A method that wins on unmodified standard benchmarks gives the reader evidence that does not depend on choices the authors made about the data. Reporting the standard and the modified datasets separately would let readers see this directly.

Evidence: out/verified_outcomes.csv (smote_kind = plain SMOTE; KNNOR's aggregate, row 295, set aside by rule R4); counts reproduced by build/reconcile_counts.py (out/reconcile_counts.json). Of the 541 rows, 3 could not be matched to a dataset entry and 15 were matched by row order only (over all 835 rows: 4 and 22). Paper-level baselines: out/full_summary.json, papers, smote_kind. RBBoost: rows 574–659; supplement Table 3 pp.3–5 and Table 11 pp.17–18; protocol p.101 and family design p.102 of the paper; review recorded in CROSS_TABLE_REVIEW

Table 6: Granularity of the Results Each Paper Reports (73 Papers)

Per-dataset table	47
Per-dataset, partial (some datasets, metrics or classifiers only)	8
Plots only	5
Averages over all datasets only	5
Ranks only	3
No comparison with other methods	5

of build/reconcile_counts.py.

1.7 Can the Reported Results Be Checked?

Of the 57 papers that use SMOTE as a baseline (44 plain, 13 wrapped), 13 give a reader no per-dataset pair of scores in the paper itself: the results are plots (Borderline-SMOTE [han2005borderline], Safe-Level-SMOTE [bunkhumpornpat2009safelevel], Cluster-SMOTE [cieslak2006cluster], k-means SMOTE [douzas2018kmeans], GLOS [han2023glos]), averages (G-SMOTE [douzas2019gsmote], CISO [yadav2025ciso], SplitBal [sun2015splitbal], QC-SMOTE [upman2026qcsmote]), or ranks (MDO, RBO [koziarski2019rbo], MC-NCLWO [tao2025mcnclwo]), and the review [hassanat2022stop] compares synthetic points rather than classifiers. A fourteenth paper, KNNOR, prints its pair of scores only as a mean over its datasets.

Baseline results are also reused across papers. QC-SMOTE copies its baseline numbers from the VS-SMOTE paper [qiu2025vssmote], and every SMOTE-all value in MC-CCR's Table 6 is identical, to two decimals and on all 20 datasets, to the SMOTE-all column of MC-RBO's Table VI, from the same group and with the same C5.0 setup. The same pattern appears across research groups: the Baseline, SSMOTE, GCS, ABNC, OSMOTE and AMDO columns of EMDO's Table 5 [yao2021emdo] repeat, value for value on all nine shared datasets, the MAUC results of AMDO's Table 6 [yang2018amdo]; EMDO states that it uses the same datasets as AMDO but not that the baseline results were copied. Reusing a baseline is efficient, but unless it is stated, a reader cannot tell whether the comparison was run under identical conditions. Several of these papers place per-dataset results on GitHub or in material outside the paper. Where only ranks are given, the paper's own table can still be informative: in RBO's Table 2 (CART, no added noise) SMOTE has the better mean G-mean rank, and CISO reports itself statistically equivalent to SMOTE.

For these papers the headline claim cannot be falsified from what is published. This does not mean the claims are wrong; it means a reader cannot tell whether an average improvement comes from consistent gains or from large gains on a few datasets.

Evidence: out/full_summary.json granularity and per-paper notes; MC-RBO Table VI p.10 vs MC-CCR Table 6 p.10 (ai-draft/full/MC-RBO.json notes). MWMOTE Table 5 audit: of 15 checkable rows, minority counts exact in 13, four rows with wrong N or majority (single paper, case only).

1.8 Recommendations for Reporting Data

1. Name the repository and version of every dataset, and whether the full set or a training file was used.

2. State which class or classes form the minority and why, in the text and not only in a table.

3. Declare every modification (binarisation, class merging, undersampling, noise, dropped attributes) in the data section.

4. Give the imbalance-ratio formula, not only its values.

5. Report per-dataset results for the proposed method and plain SMOTE, and report standard and modified datasets separately.

6. Separate tuning datasets from evaluation datasets, as CISO and LS [CIESLAK2008LS] do, and run the baselines rather than copying their numbers.

7. Report at least one measure of where the minority examples lie (for example the shares of safe, borderline, rare and outlier examples) next to the imbalance ratio.

[Toon] Keep, cut or reword; each item should point back to a finding above.